

Report on Developing a FinTech App

KYC Assistant

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1. Background and scope

The following report outlines the stages involved in developing a FinTech app prototype, from gathering initial requirements and creating the product vision, to delivering the front-end prototype and planning future product features. The report details how agile project management methodologies have been applied to the development of the application in its design to implementation.

1.1. Business opportunity & problem statement

During the discovery phase of the project, research into the FinTech and Compliance space showed that the cost of compliance is a growing challenge for businesses in both regulated and non-regulated sectors ([PwC US, 2024](#)). LexisNexis reports that UK financial institutions alone spent £38.3 billion on financial crime compliance (FCC) in 2023 - representing a 12% increase from 2022 and a 32% rise since 2021. They also note that, on average, UK financial services firms spend £21,000 per hour on FCC ([LexisNexis Risk Solutions, 2024](#)).

These rising costs place pressure on business margins but, if regulations are ignored or FCC practices are improperly managed, non-compliance can result in severe penalties. For example, in October 2024, Toronto Dominion (TD) Bank agreed to pay over \$3 billion in penalties for failing to maintain adequate financial crime controls, which facilitated money laundering through its institution ([Sherman, 2024](#)).

This report looks to address some of these pain points by proposing an app that shifts away from costly, manual compliance processes towards an automated, continuous compliance model ([Britsch, 2024](#)). The app aims to support teams and businesses in enhancing efficiency, reducing costs and combating financial crime.

2. Requirements gathering

The first phase of the project was to gather requirements for a potential FCC app by interviewing stakeholders and observing processes such that user needs and business solutions could be identified.

The requirements are set out in subsequent functional and non-functional requirements which are converted into user stories.

2.1. Stakeholder interviews and observing processes

Stakeholder interviews were undertaken with a sub-set of an FCC team specialising in Anti-Money Laundering (AML). The interviewees included KYC (Know Your Customer) analysts, a KYC team lead and a team Director, specialising in AML and project delivery. Key takeaways from the interviews are set out below (detailed interview notes are set out in Appendix A):

Current pain points and challenges:

- **Manual processes:** Excessive time spent copying/pasting data, higher error risk, multiple review stages, and less time for risk analysis and skill development. Analysts manually compile data

from usually from one website, UK Companies House¹ (though multiple other sources can also be used) including financial statements and public records, resulting in repetitive, time-consuming tasks.

- **Ineffective tools:** Previous systems have been overly complex, had unnecessary features, frequent breakdowns, and were eventually decommissioned.
- **Document formatting and reporting:** Inconsistent formatting across analyst reports requires time-consuming reviews and rework, increasing costs.

Observing and testing sub-processes

A manual sub-process was explained to the team and was reperformed to see how long it takes for someone new to run from end to end. The process included searching for a company on UK Companies House (Companies House), copying and pasting data into a template, summarising the nature of business of the company and extracting recent filings from the document repository.

Reperforming the copying and pasting process by a new, untrained team member could take up 20 minutes. This time taken increased to at least an hour where the nature of business summary is included.

Features and functionalities desired from a KYC platform:

- Automated data extraction from Companies House.
- User-friendly design with only essential features to avoid complexity.
- Export of formatted, auto-generated KYC reports.
- Centralised repository for Company records.
- Automated company nature of business summaries.

2.2. Functional requirements summary

The features and capabilities that the app could have, as informed by user interviews, process observations, and competitor analysis, are outlined in the table below.

¹ UK Companies House incorporates and dissolve limited companies. They register company information and make it available to the public (House, 2024).

ID	Requirement	Description	Priority
F1	Company search	Users can search for businesses in Companies House by name or registration number.	High
F2	Company profile view	Display company details such as name, status, addresses and Directors.	High
F3	Company profile download	Ability to download the summary of company data to excel or PDF.	Medium
F4	Document download	Ability to download recent company reports and filings.	Low
F5	User authentication	Users log in must be authenticated.	High
F6	Company portfolio	Ability to save user searches of companies into a portfolio view for future reference.	High
F7	Audit trail	Log all user actions and track changes for quality and audit purposes.	High
F8	Notifications	Send alerts on changes to company data for tagged companies.	Medium
F9	Search history	Maintain log of previous searches so users can revisit their history.	Low
F10	Nature of business	Automatically generate a summary of the company nature of business per Companies House data.	Medium
F11	Red flags	Provide details of AML (or other Financial Crime) red flags based on Companies House data.	Medium

Table 1. Functional requirements

2.3. Non-functional requirements summary

The non-functional requirements that the app could have, as informed by user interviews, process observations, and competitor analysis, are outlined in the table below.

ID	Requirement	Description	Priority
NF1	Scalability	Support at least 1,000 concurrent users without a degradation in performance.	Medium
NF2	Performance	The system must return search results within 2 seconds for 90% of searches and load company profiles in 3 seconds.	High
NF3	Data loss	Ensure no data loss in the event of system crashes or unexpected shutdowns.	High
NF4	Error handling	System errors should be caught and logged, and users should see a clear, user-friendly error message.	Medium
NF5	Data Privacy	Ensure full compliance with GDPR and other privacy regulations, supporting user requests for data erasure.	Medium
NF6	Availability	Ensure 99.9% system uptime, excluding scheduled maintenance, and enable automatic failover within 30 seconds of a failure.	Medium
NF7	Auditability	Log all user actions and changes for audit trail purposes, ensuring the logs cannot be tampered with and are retained for 7 years.	High
NF8	Data Backup	Daily data backups must be automated, and full data recovery should be possible within 24 hours of a catastrophic event.	Medium
NF9	Compliance	Ensure compliance with AML regulations, GDPR, and ISO 27001 standards for security, privacy, and encryption.	Medium
NF10	Secure access	Enforce strong password policies to increase platform access security.	High

Table 2. Non-functional requirements

2.4. User stories

The research, along with functional and non-functional requirements, has been used to create user stories. These user stories are mapped to corresponding app features and acceptance criteria, which will guide product delivery and help assess when a user story is complete. The user stories will be prioritised in the subsequent phases of the project and addressed in later sections of the report (under “Project Management”).

No.	Feature name	User story	Acceptance criteria
US1	Search companies	"As a KYC analyst, I want to automatically retrieve company details from Companies House when I enter a company name or number, so that I can view businesses details."	- The app should retrieve the company name, registration number, status, incorporation date data within 10 seconds of input for 90% of searches using direct Companies House API integration and display this in the app. - The app should display matching companies for partial name searches.
US2	View company data	"As a KYC analyst, I want to view comprehensive company information (including company name, status, addresses and Directors) after performing a search, so that I can efficiently	System must display key company details including name, status, registration number, address, officers, and filings.

		assess the company's profile and risk factors."	
US3	Export company data	"As a KYC analyst, I want to download company details as an excel or PDF, so that I can eliminate manual copy-paste tasks and reduce processing time."	The app should have an export function that extracts and consistently formats all key company data points.
US4	Export filings	"As a KYC analyst, I want to view and download the most recent financial statements and filings from Companies House."	- The app should have an export function that downloads any documentation from the last year on Companies House. - The app should display filed documents chronologically and enable the export of multiple documents in their original format with a single click.
US5	Portfolio view	"As a KYC analyst, I want to save and organise my company searches into customisable portfolio views, so that I can efficiently monitor multiple companies."	- Users can save individual company searches to one or more portfolios. - Users can create, name, and manage multiple portfolios.
US6	Authentication	"As a KYC team lead, I want to ensure that the system has strong authentication mechanisms, so that only authorised users access the system."	- Users must authenticate with strong passwords meeting industry standards. - System must support password reset functionality.
US7	Audit history	"As a Compliance Auditor, I want a complete digital trail of data collected, so that I can demonstrate the source, timestamp and accuracy of all gathered information during reviews."	- The app should maintain a complete digital trail of all data access, changes, and sources with timestamps. - Data should be accessible and stored for up to 7 years.
US8	Business summary	"As a KYC analyst, I want to automate the drafting of the nature of business summary, so that I can reduce the time spent manually explaining what a business does."	The app should generate standardised business descriptions from SIC codes, along with extracting key data points from financial statements metrics such as revenue/ profit, while allowing manual edits.
US9	Search history	"As a KYC analyst, I want to access and manage my search history, so that I can easily track and revisit previously investigated companies without duplicating work."	- The system must maintain a complete chronological log of all user searches. - Search history must include timestamp, search terms, and search filters used. - Users can filter search history by date range, company name, or search type.
US10	Update notifications	"As a KYC analyst, I want to be notified when there are changes to company information at Companies House, so that I can	The app should enable monitoring of companies simultaneously with a status dashboard and bulk action capabilities.

		promptly update our due diligence and risk assessments.”	The app should notify users within 1 hour of Companies House updates, specifying changed information.
US11	Red flags/ risks	"As a KYC analyst, I want the system to automatically identify and alert me to potential AML and Financial Crime red flags based on Companies House data, so that I can quickly see where risks might lie and conduct enhanced due diligence where necessary."	- System must automatically check for and flag key risk indicators². - Red flags must contain a timestamp of detection and a history of all identified red flags.
US12	Scalability	"As a KYC team lead, I want the system to handle high concurrent usage during peak periods, so that our team can work efficiently."	- System must support 1,000 concurrent users. - Company profiles must load within 3 seconds. - No degradation in performance during peak usage.
US13	Data backup	"As a KYC team lead, I want automated backup and recovery procedures, so that we can quickly restore service after any disruption."	- Daily automated backups must be performed. - Full data recovery must be possible within 24 hours. - Backup success/failure must be logged and monitored.
US14	Error messages	"As a KYC analyst, I want to receive clear error messages when issues occur, so that I can understand and report problems effectively."	- Error messages must be clear and user-friendly with users being notified of system-wide issues. - Technical error details must be logged for support.
US15	Data protection	"As a Data Protection Officer, I want to ensure comprehensive GDPR compliance in our data handling processes, so that we protect individual privacy rights and meet regulatory obligations."	- System must support data subject access requests. - Personal data must be encrypted at rest and in transit. - System must support data erasure requests.
US16	Compliance alignment	"As a Compliance Auditor, I want to ensure the system meets all relevant security and regulatory standards, so that we maintain compliance with AML regulations and ISO 27001 requirements."	- System must implement core ISO 27001 controls including access control, encryption, and security monitoring. - System must demonstrate compliance with AML regulations and user company policies and procedures. - System must maintain comprehensive audit trails and documentation for all compliance activities.

Table 3. User stories

² Some examples of red flags that could be identified from Companies house data are as follows: Frequent changes in company directors or ownership, connections to high-risk jurisdictions, multiple companies registered at the same address, individuals being a Director of multiple companies, dormant company suddenly becoming active, missing or late financial filings, company with high-risk industry SIC code etc.

3. Product Vision

Before the team started building the product prototype, a product vision was defined to focus the team and drive them forward in working towards shared goal by capturing in a few sentences what the app is trying to achieve ([Pichler, 2024](#)).

Roman Pichler offers an approach to defining the product vision through a Product Vision board ([Pichler, 2022](#)). The Product Vision board for our app is set out below:

Product Vision Board			
Vision: • Reduce the time spent by teams on repetitive, manual data entry tasks so that they can spend more time on more complex analysis. • Transforming compliance into action - automating KYC to free teams for meaningful work while protecting financial integrity.			
Target group <u>Customers:</u> Financial and non-financial institutions that conduct anti-money laundering due diligence Professional services firms with large client portfolios Compliance departments in regulated industries <u>Users:</u> 1 st and 2 nd Line of Defence including businesses, financial crime and compliance teams.	Needs Reduce time taken to conduct manual, routine KYC checks. Reduce human error and data formatting inconsistencies Audit trail and adherence to policies and regulations.	Product KYC Data Assistant streamlines regulatory compliance by extracting and organising UK Companies House data into actionable business intelligence. The solution automates routine compliance checks and transforms real-time raw company data into structured formats, enabling rapid customer due diligence and risk assessment.	Business Goals 50% reduction in KYC processing time Standardises KYC documentation across the organisation and reduces human error Scales efficiently with business growth

Table 4. Product vision board

In summary, the product vision for the app is as follows:

KYC Assistant: Automated compliance for time-pressed teams.

4. Project Management

Agile tool & Methodology

The project will be run using the Scrum methodology ([Scrum.org, 2024](https://www.scrum.org/)), using one week sprints in order to quickly iterate from Minimum Viable Product (MVP) to user-tested prototypes (V1 and V2).

The project length is four weeks, which has been divided into a planning and research week, plus three subsequent sprint weeks:

- Week one is focused on gathering user stories, building the first set of front-end designs, and deciding on the correct architecture for our application. Product discovery will be a focus of this week, to ensure that our understanding of the problem statement is correct (note results of week one have been set out in earlier sections of this report).
- The second week, Sprint 1, will be focused on shipping the MVP and is primarily aimed at getting a simple, value-adding product into users hands to gather user feedback.
- The third week, Sprint 2, will be focused on shipping Version 1 (V1), an iteration on the MVP defined by the user feedback gathered as a result of that release. Key activities will be as follows:
 - Review customer feedback
 - Map customer journeys
 - Scope and design features
 - Building time
- The final week, Sprint 3, will iterate further on the V1 build, again incorporating additional feedback from our ongoing user interviews.

For the purposes of this report, we will focus on the delivery of the front-end product.

Weekly meeting cadence

	Monday	Tuesday	Wednesday	Thursday	Friday	
Main meeting	Sprint planning	Daily scrum	Daily scrum	Daily scrum	Demo	Retrospective
Purpose	Agree the tasks and output for the week	Check in on the team. What did everyone complete yesterday? What is everyone working on today? What are the blockers?			Demo to the wider company what has been built for internal feedback	What went well in the last sprint? What could be improved in the next sprint? What issues arose and how can they be solved if not already? Review product backlog in advance of next sprint
Attendees	Product Owner Scrum Master Lead Engineer Lead Designer	Scrum Master All engineers			Full team	Full team

Table 5. Weekly meeting cadence

Feature prioritisation / Decision making

The Product Owner will use RICE scoring to determine which are the features where we have the greatest impact on our customers, while having to do the least engineering possible. To avoid decisions made by “gut feeling” or a lack of direction in the team, the RICE scoring matrix will determine what features will be added to the product roadmap, ensuring the team is able to ship quickly and gather customer feedback rapidly. This should help to reduce abortive work.

Features suggested by customers which are subsequently deprioritised after RICE scoring will be stored on a Product Backlog for future review.

Below are the RICE scored features we have determined for the early versions of our application:

No.	Feature name	Reach	Impact	Confidence	Effort	Score ³	MOSCOW
US1	Search companies	10	5	8	3	133	Must have
US2	View company data	10	8	8	5	128	Must have
US3	Export company data	9	8	8	6	96	Must have
US4	Export filings	6	5	5	8	19	Won't have
US5	Portfolio view	9	8	7	7	72	Must have
US6	Authentication	10	4	9	5	72	Must have
US7	Audit history	6	9	7	6	63	Must have
US8	Business summary	8	7	8	8	56	Should have
US9	Search history	7	4	7	3	65	Must have
US10	Update notifications	6	8	6	9	32	Could have
US11	Red flags/ risks	7	9	7	9	49	Could have
US12	Scalability	10	10	4	7	57	Should have
US13	Data backup	9	3	5	7	19	Could have
US14	Error messages	8	4	5	6	27	Could have
US15	Data protection alignment	10	2	5	6	17	Should have
US16	Compliance alignment	10	2	5	6	17	Should have

Table 6. RICE scored features and user stories

³ RICE score calculated as Reach x Impact x Confidence / Effort. Each category is calculated out of 10 with the score being the output of the calculation.

Per the above table, the following features have been prioritised to be prototyped over the next three sprints with the remaining features and user stories being added to the product backlog:

- Search companies (US1)
- View company data (US2)
- Export company data (US3)
- Portfolio view (US5)
- Authentication (US6)
- Audit history (US7)
- Search history (US9)

Please see Appendix B for the full proposed product roadmap.

Feature sizing / estimation

The project's Lead Engineer will be accountable for estimating the build time and testing time required for features. Our estimation technique will be t-shirt sizing, with an understanding that as these features are new and the complexity is unknown, it's important to be conservative in our estimates to ensure sufficient time for testing each new feature in the application.

In addition, due to the relatively short length of sprints, and the tight overall project timeline, we will mostly prioritise features which fall into the "Small" or "Extra Small" effort categories, to reduce the risk of over-committing to a build that's more complex than necessary.

5. UML Diagrams

5.1. Use case diagram

The KYC app supports KYC analysts, KYC team leads, and Compliance Auditors in performing core system activities related to searching for companies, viewing company details and exporting company information. The system must provide access to essential company data, export functions, audit trails, and portfolios.

The system includes four key actors who interact with various functionalities: the KYC Analyst, KYC team lead, Compliance Auditor, and the external Companies House API system. The KYC Analyst and KYC team lead, for the purposes of the prototype, can perform the same actions and so they are generalised into an abstract KYC team member.

The Compliance Auditor focuses specifically on audit history functionality, monitoring and reviewing the system's audit trails. This audit system is extended by the authentication process, ensuring that all login attempts, access controls, and security measures are properly logged and traceable.

Several key relationships exist between the different use cases with most use cases being included within the view company data use case. The portfolio system includes the search companies functionality, as users need to search for companies before adding them to their portfolios. All functionalities operate within the defined KYC Assistant system, with only the Companies House API as an external entity.

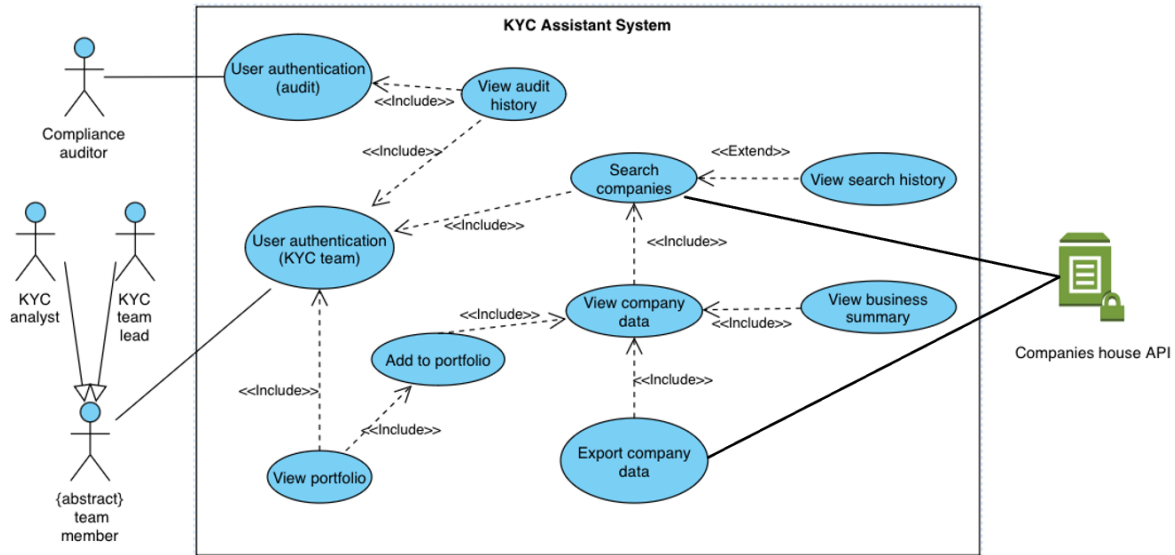


Figure 1. Use case diagram

5.2. Class diagram

The UML class diagram models the systems core components. The diagram⁴ also includes key entities, attributes and their relationships that enable core KYC Assistant functions to run including company search, company data retrieval (through the Companies House API) and exporting data.

The system combines the KYC analysts and KYC team leads into one class, KYC User, who is able to search for companies via the Companies House API and manages portfolios. Compliance Auditors are also a class who are able to review audit histories. All users can view audit histories. In future iterations, the system could have role-based controls so that KYC team leads have different access rights. User authentication is handled by an Authentication Service class to ensure secure access management, while company data retrieval is undertaken through the Companies House API class.

The system's main workflow involves KYC Users searching for and monitoring companies through the Companies House API, which fetches official company data including officers, filings, status and address information. Additional services also include system settings, and error logging. Future iterations of the product would use data called from the Companies house API to identify red flags and the nature of business summaries (these have again been added to the product backlog).

⁴ The diagram has been modelled using Mermaid Live Editor (2024).

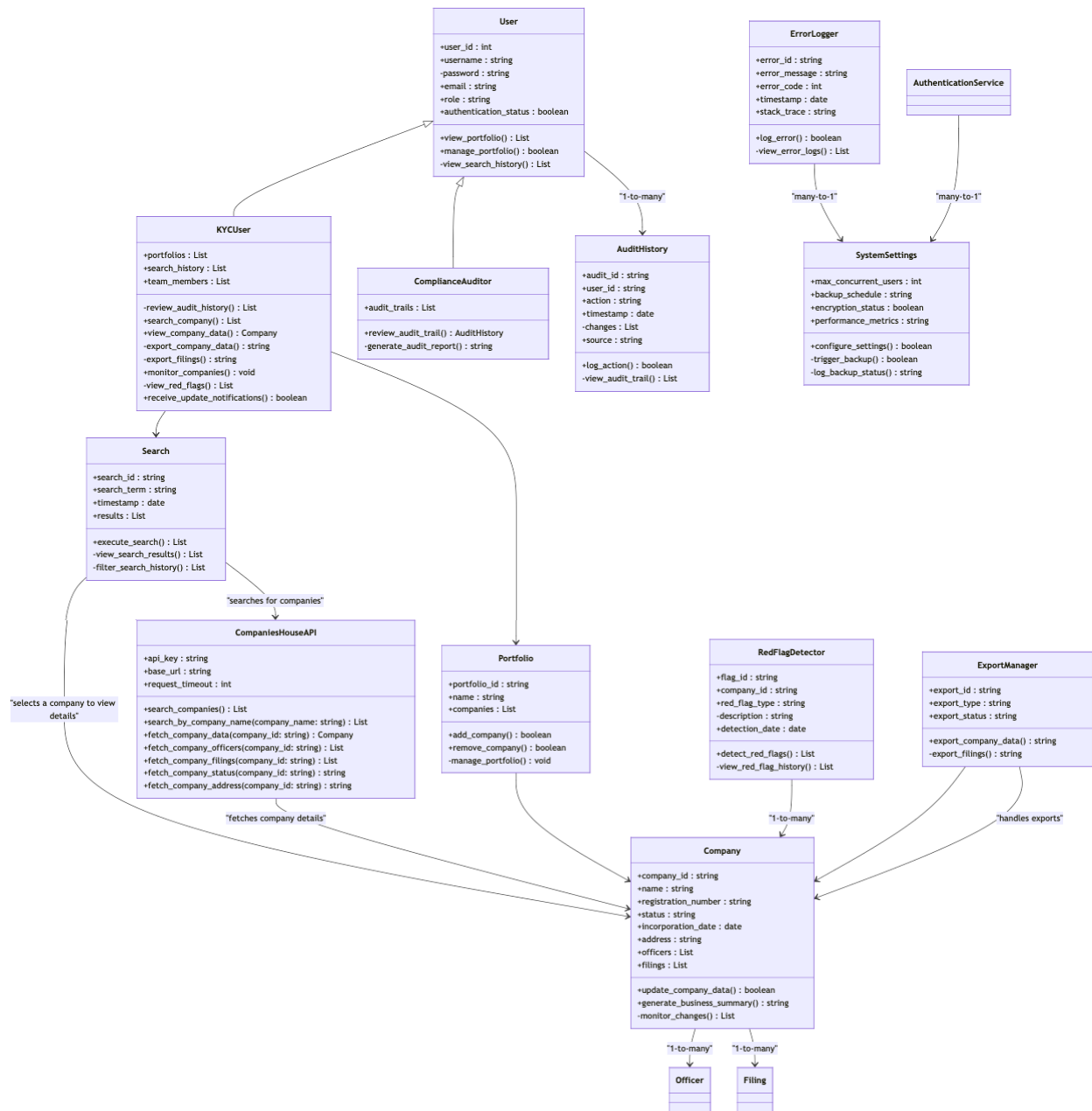


Figure 2. Class diagram

5.3. Sequence diagram

The sequence diagram⁵ illustrates the workflow of KYC Assistant focusing on the sequence of events for processes and how different components interact to process and manage company information.

⁵ The diagram has been modelled using Mermaid Live Editor (2024).

The diagram begins with the authentication process, where a user accesses the system through the user interface. The Authentication Module verifies their credentials establishing a secure session for the user to proceed with viewing audit histories or searching for companies.

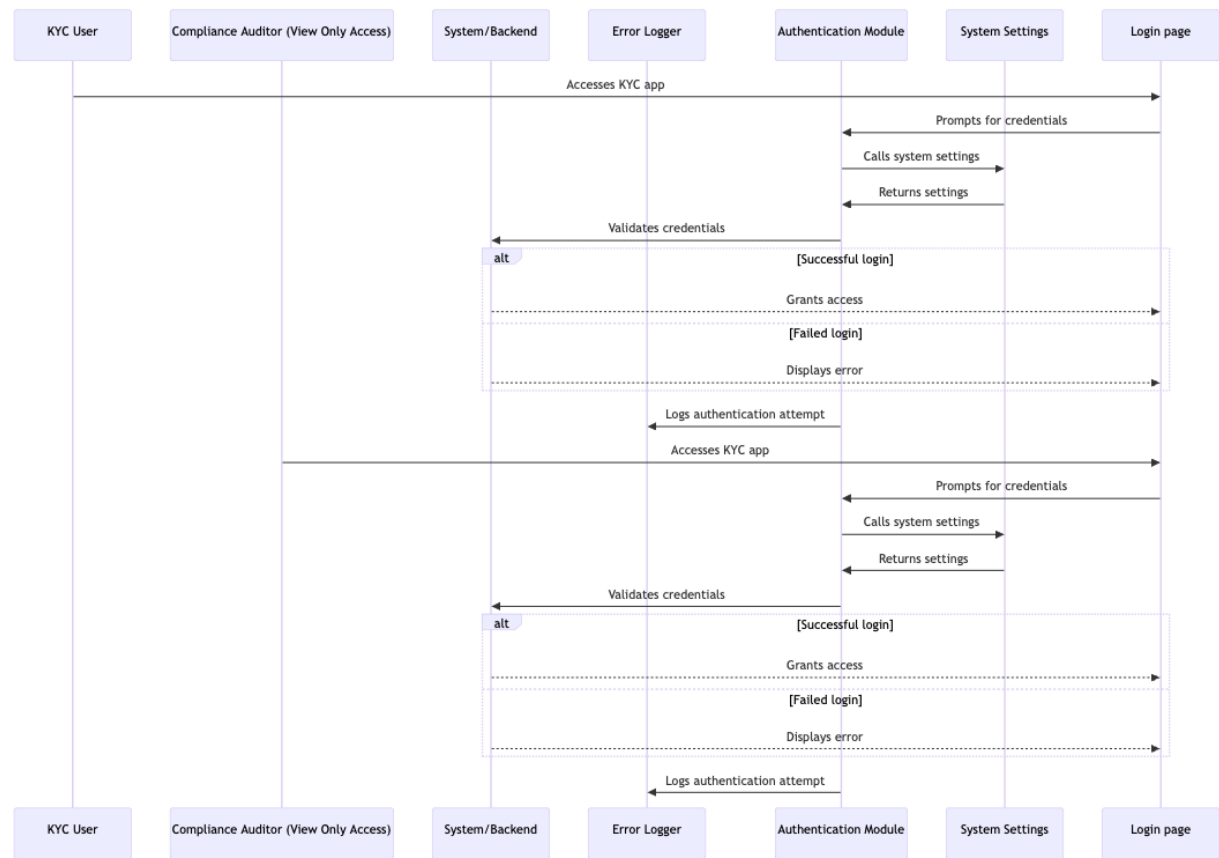


Figure 3. Login sequence diagram

Once authenticated, the core functionality revolves around company searches and data management. When a user initiates a company search, the request flows from the UI through the system to the Companies House API.

The portfolio management aspect is represented in the following sequence, where users can save and organise company information. When a user selects a company to save, the Portfolio system handles the storage of this information, and all actions are logged by the Audit History for compliance and tracking purposes.

The diagram also shows the export functionality, where users can generate reports through the Export Manager. This process demonstrates how the system formats and delivers company data in downloadable formats while maintaining proper audit trails of all export activities.

A key feature illustrated is the system's real-time update capability. The diagram shows how the Companies House API can be called to ensures users always have access to the most current company data.

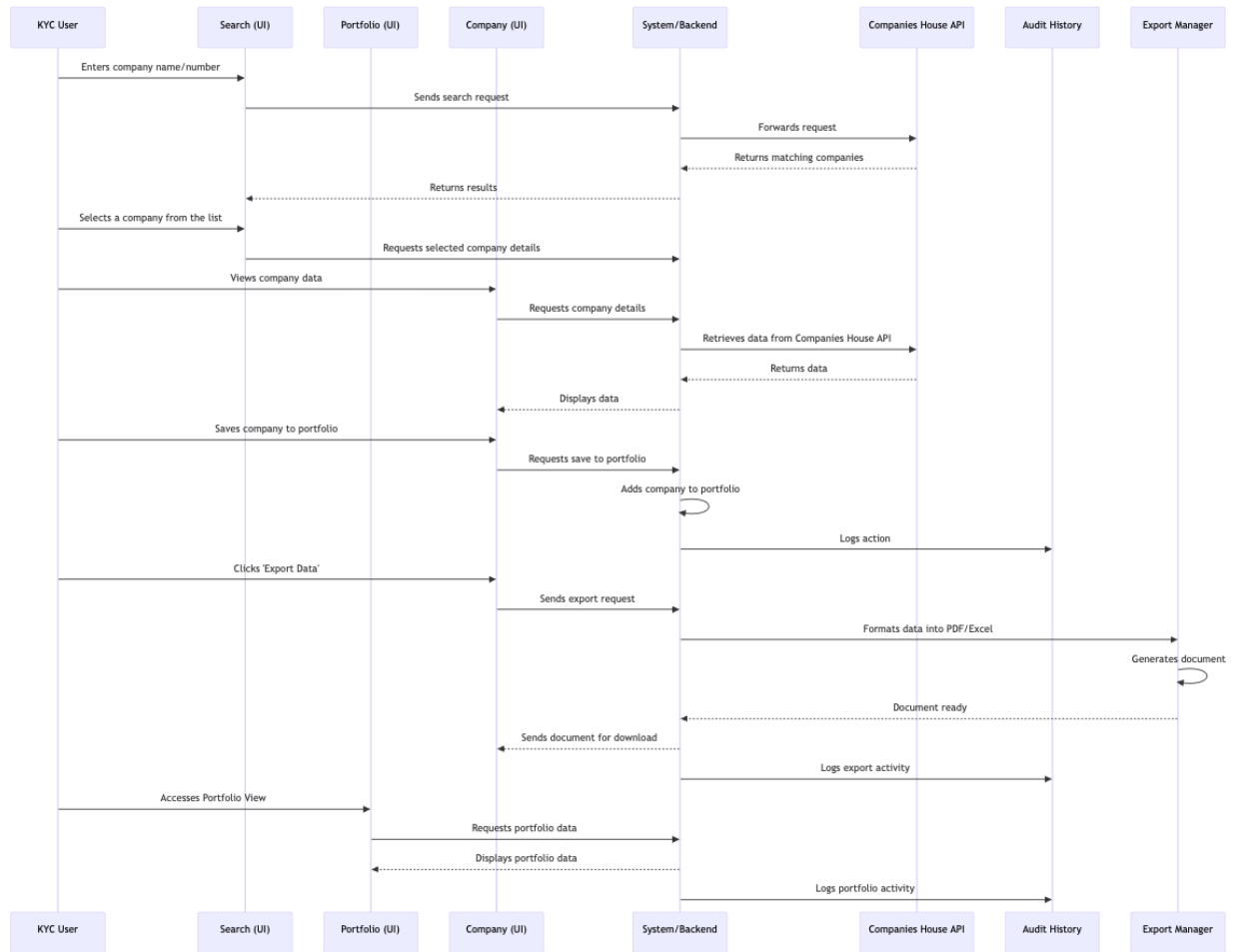


Figure 4. Search, view company and export data sequence diagram

6. Prototype

6.1. Prototype feedback

Wireframes for the prioritised user stories and features⁶ in the front-end system were developed using UX Pilot ([UX Pilot, 2024](#)) in the first sprint week and are set out in Appendix C.

Feedback from Sprint 2 and 3:

As set out in the agile project management plan, every Friday a demo will be run to showcase what has been developed that week to obtain feedback for future sprints and prototypes. The table set out in Appendix D contains detailed feedback received on the high-fidelity wireframes however some key feedback implemented includes:

- Google and Microsoft login options have been deferred to the product backlog.
- Introduce a more prominent "Sign up" button on the login page.
- The Company details page will display more company data, such as office address, company type, Directors, and SIC code.
- It was noted that a search results page between the search and company details pages had not been prototyped and so has been included in the final prototype below.

6.2. Prototype

The prototype has been hosted on the PythonAnywhere system ([pythonanywhere.com, 2024](#)) and provides a live prototype for users to test and interact with the link is here <https://sharris2011.pythonanywhere.com/login>,

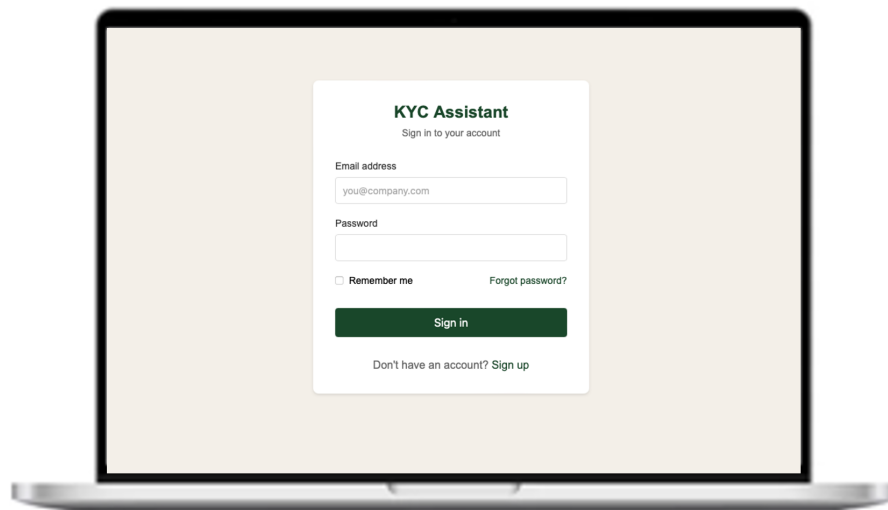


Figure 5. Login screen

⁶ Laptop outline from Adobe Stock ([Adobe Stock, 2024](#))

The users first interaction with the application begins with a clean, minimalist login screen featuring the KYC Assistant branding, basic email and password fields, and a clearly identifiable “Login” button.

There is an option to enable “Remember Me,” which keeps the user logged in for future sessions. To assist users who may forget their credentials, a “Forgot Password” link is provided for account recovery. Additionally, there is a sign-up link that encourages new users to create an account, ensuring smooth onboarding for those accessing the application for the first time. Note that these functionalities are currently on the product backlog, however users can log into the system using the username, TESTUSER@gmail.com, and password, Password1.

Strengthening user access and authentication security in future iterations has been added to the immediate next functionalities on the product roadmap.

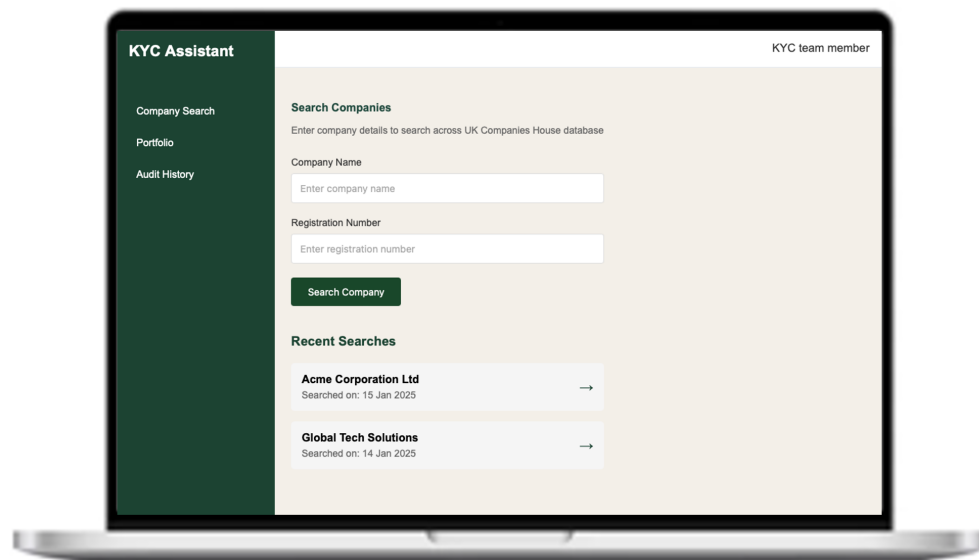


Figure 6. Home view and company search screen

Once logged in, there is a consistent interface design across all screens. A dark green navigation sidebar that provides quick access to the three core functions “Company Search”, “Portfolio” and “Audit History”.

As the primary functionality of the app, the “Search Companies” interface acts as the home page and streamlines users’ ability to finding company information, featuring two primary search fields for company name and registration number. This screen also contains a "Recent Searches" section, allowing users to quickly return to previously viewed company searches, enhancing workflow efficiency and user experience.

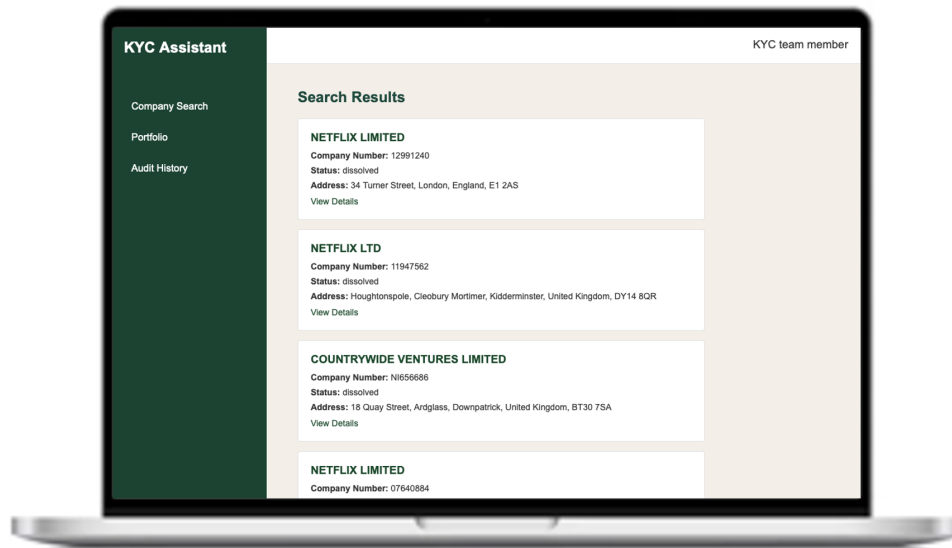


Figure 7. Search results screen

After searching for the company using either the company name or ID, the app shows possible matches on the name or company ID search, listed by most relevant.

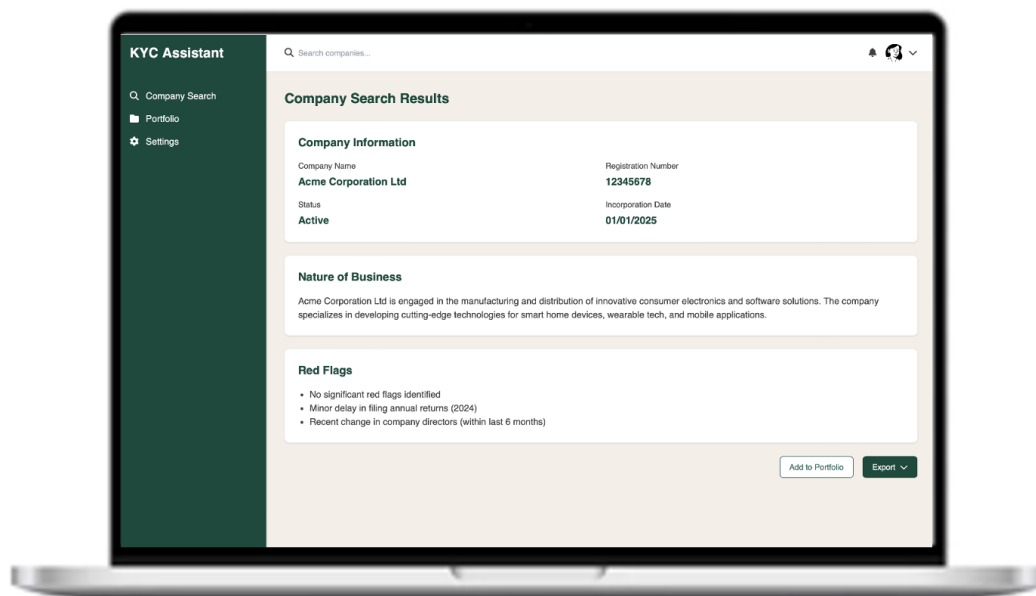


Figure 8. Company details screen

The company details screen provides detailed information about a company after a successful search query. At the top, the Company Information section highlights core company such as the company name, registration number, status (e.g., active), and incorporation date.

Below, the Nature of Business section contains a short description of the company's operations, providing users with insight into its primary activities and areas of focus. Additionally, this view contains a Red

Flags section, which will highlight potential risks associated with the company from Companies House data. These features are being added to the product backlog for medium-term implementation.

To help users save and export their search results, there are two functional buttons: “Add to Portfolio” and “Export”. The “Add to Portfolio” button allows users to save the company for further assessment within their portfolio, while the “Export” option lets users download the company’s data as a PDF.

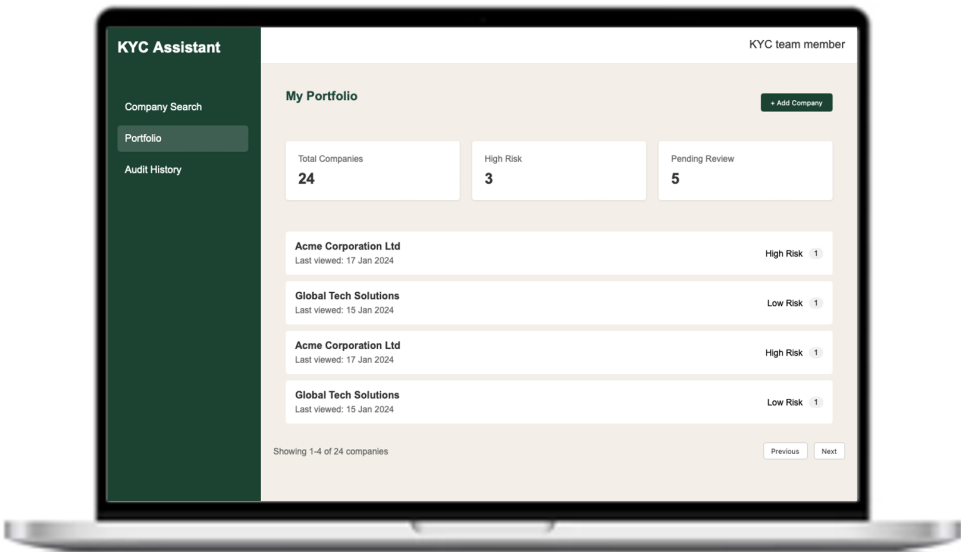


Figure 9. Portfolio view screen

The My Portfolio section functions as a central dashboard, offering users visibility of companies within their review and workflow. It displays metrics including total companies within the portfolio, higher risk companies (to be drawn from the red flag section of the company details screen).

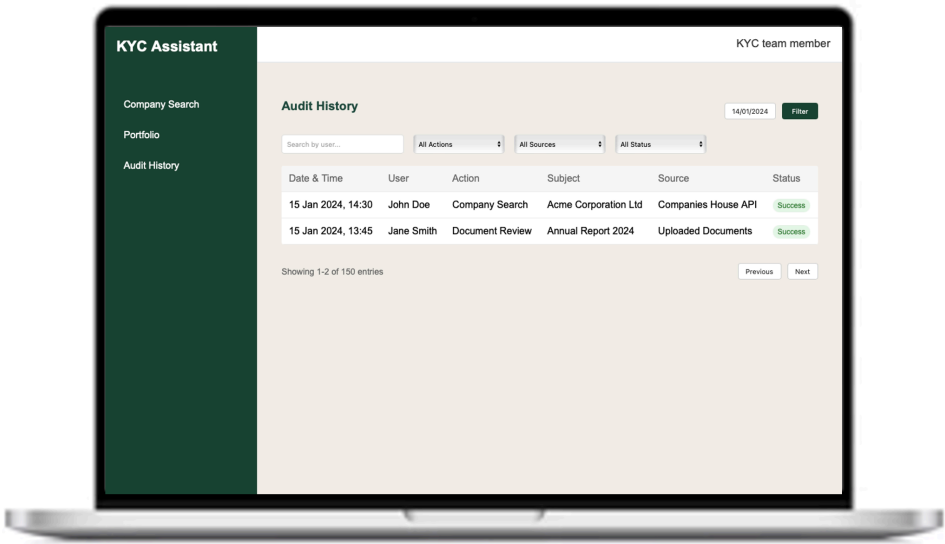


Figure 10. Audit history screen

The final screen showcases the Audit History functionality, primarily for compliance tracking and accountability. This interface provides filtering options for dates, activity types, and statuses, allowing users to maintain detailed records of all compliance-related actions and reviews.

Currently, the view is accessible to all users and roles as, in the short term, it would be useful for KYC Analysts and team leads to see what actions have been performed in early stages of rollout. As the app matures, additional roles will likely be created for users that can view audit trails but not customer data, for example. These operations have been added to the backlog.

In conclusion, the KYC Assistant app provides a streamlined solution for managing KYC processes. It provides users with tools to search for company details and maintain an organised portfolio of monitored companies. By automating data retrieval and presenting it in a structured manner, the application reduces the manual effort involved in KYC processes and ensures compliance officers have access to accurate, up-to-date information at all times.

7. Prototype testing

The testing strategy for the KYC app focuses on automation, security, and regulatory compliance. The future testing strategy will look to embed tests directly into the codebase for efficient testing of the end-to-end platform and an integrated part of the apps development.

Unit testing – Companies house API

Unit tests are written in code using a testing framework like pytest ([pytest.org, 2024](https://docs.pytest.org/en/latest/)) to ensure each function operates as intended (Destefanis, 2024).

For example, we would look to run tests for login authentication, Companies House API calls, data export functions, and search functionalities.

Integration testing

Integration testing ensures that interconnected modules, such as API calls and search functions, work together as intended (Destefanis, 2024). The team would run these tests to ensure that modules work together as expected. For example, testing the integration between the app and Companies House API.

Security testing

Evaluates system defences against unauthorised access and vulnerabilities, (Destefanis, 2024). These tests would be run once the password and authentication functionalities have been built out and continuously run to ensure that data is only accessible and alterable by authorised users.

Performance testing

These tests assess the app's ability to handle up to 1,000 concurrent users, ensuring fast load times and responsiveness, and would use tools that would simulating peak loads (Destefanis, 2024). The tests would ensure that key metrics such as time to load/ download/ view are within expected timeframes.

Audit history

Lastly, the team would verify that all user actions, including searches, exports, and edits, are recorded with timestamps and user details in the Audit History functionality for accountability and traceability.

8. Product roadmap and next iterations

Going forwards, the team would look to continue to build on the current app prototype with additional features from both the original user story list but also features added to the product backlog over the course of developing the app.

The roadmap below is broken down into deployment priorities for the next 3, 6 and 12 months to focus on features that would provide immediate business and user value while building the core system functionalities to enable scalability and future growth. These functions would be developed in the same way as the current prototype, using a scrum methodology as set out in the “Project Management” section of the report.

Immediate Priority (0-3 months):

The team would look to focus on the apps core infrastructure and building out essential features as identified from the requirements gathering stage.

- **Log in and authentication** – the app currently has log-in capabilities however it would benefit from deploying reset password and sign-up functions, along with more robust authentication mechanisms (such as a single-sign-on (SSO) for corporate users). Google and Microsoft logins could also be included to reduce friction and enhance security.
- **Role-based access** - we would look to implement role-based access controls for KYC team leads, as this would enhance the security of the system and enable future rollout of KYC team specific functionalities.
- **Sharing of portfolios** – with the above point in mind, KYC analysts could have an option to share portfolios with their team leads or with other analysts for quick access to previously saved search portfolios.

Medium Term (3-6 months):

- **AI-powered nature of business summaries** – as noted in the requirements gathering phase of the project, an AI-powered nature of business summary that leverages the Companies House API would provide a boost to team efficiency through reducing time taken to draft these summaries. The Scrum Master gave this function a high effort score in the RICE scoring and so would likely require one or more sprints to fully develop and test.

Long Term (6-12 months):

- **Cloud hosting** - As the system has already been deployed, the team would also look to host it on a cloud platform like AWS or Azure for scalability. In addition, we would look to implement a microservices architecture to support modular growth.
- **Additional API integration** - The app could also be expanded to integrate with other, international publicly available information/ registry information to enhance automation. Connect to international company registries, starting with major markets (EU, US, Singapore) based on customer demand.
- **Integration with other AML systems** – The app could also integrate with other company systems such as CRMs, risk assessment, transaction monitoring or sanctions screening systems to provide an end-to-end compliance workflow and move towards continuous compliance model.

9. Appendices

Appendix A – Stakeholder Interview Notes

Stakeholder Interview Notes: KYC Director and AML expert

Key Responsibilities of KYC Team Director

- Oversee and manage the KYC team to ensure compliance with Anti-Money Laundering (AML) regulations and internal policies.
- Implement tools, systems, and workflows that enable teams to work effectively but to a high quality to minimize costs and help team members develop.

Current Pain Points & Challenges

- **Over-Reliance on Manual Data Processes** - Team members spend excessive amounts time copying and pasting company information from Companies House into Excel or internal templates. Analysts are focused on manual data entry instead of conducting insightful analysis on customer risk.
 - Increased potential for human error when copying and pasting information.
 - Frustration from analysts who feel they are doing data entry instead of critical analysis.
- **Ineffective tools** - A previous system was introduced to automate the retrieval of Companies House data, but it failed to deliver as expected.
 - The system was overly complex and had too many unnecessary features that distracted from the core business need meaning that the team struggled to use it.
 - Frequent technical issues and breakdowns that caused delays and disrupted workflows.
 - Ultimately, the system was scrapped as it became more of a burden than a solution.

Core features of a platform

- Automated extraction of company information from Companies House into KYC workflows to reduce manual data entry time, reduce human error and enable analysts to focus on value-added analysis rather than data entry.
- A system that is easy to use with only essential features required for KYC processes to avoid the mistakes of the previous overly complex system. Ensure the system is intuitive, with minimal training required for analysts and that it is stable and reliable to prevent downtime or system failures.
- A central repository for all company information and KYC records to access past records.
- Automated generation of KYC reports with customizable formatting options. Reduce the time analysts spend on formatting and report production. Ensure consistent reporting standards and branding.

"Nice to Have" Features

- Real-time notifications and alerts where there are changes in Companies House records.
- Simple red flag identification system based on Companies House data to support early risk assessment/ due diligence.
- Integration with existing compliance platforms, CRMs, and case management tools.

Stakeholder Interview Notes: KYC Analyst

Key Responsibilities of KYC Analyst

- Conduct customer due diligence (CDD) for new and existing clients to ensure compliance with Anti-Money Laundering (AML) regulations and internal policies.
- Document findings and maintain accurate client records for audits and regulatory reporting.

Current Pain Points & Challenges

- **Manual data extraction** - Analysts spend significant amounts of time manually copying and pasting information from Companies House into Excel or internal templates. It is a time-consuming, repetitive process and there is no automated method to transfer company information from Companies House to internal systems.
- **Nature of business summaries** - Analysts review Companies House filings to determine the company's Nature of Business.
 - The summary requires analysts to draw information from several sources including previous financial statement filings (for turnover, number of offices, employees etc.), other publicly available information sources along with the Compliance team/ Clients, requiring additional time for cross-referencing.
- **Document Formatting & Report Creation** - Analysts must create formal KYC reports that include company information, ownership structure, and risk assessments.
 - Formatting issues arise when copying data from Companies House or other sources into reporting templates.
 - Consistency issues due to multiple analysts having different formatting practices.

Core features of a platform

- Ability to automatically pull data from Companies House directly into the KYC platform or into pre-built templates to reduce manual data entry time.
- A built-in tool that can summarise the company nature of business automatically. Reduce the time spent on research and back-and-forth validation with multiple sources, the compliance team and clients.
- Automated generation of KYC reports in a consistent format, regardless of who is preparing them. One-click report generation with minimal human intervention.
- Single source of truth for all company records, allowing for easier updates and tracking of changes over time. Central location for tracking and referencing previous client information.

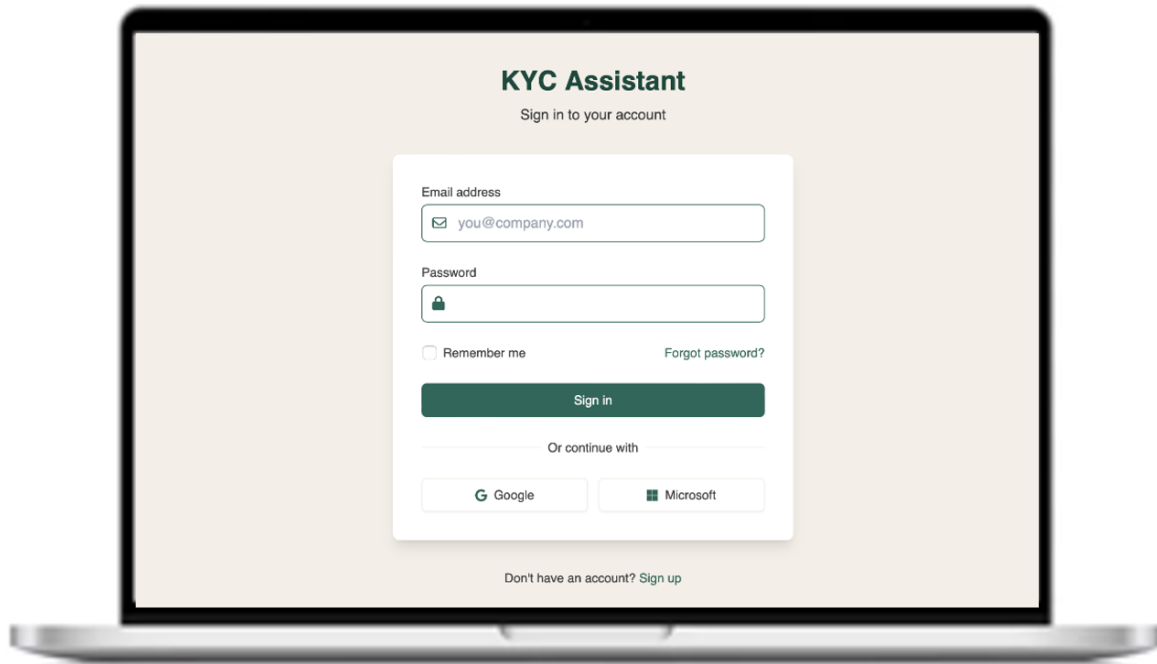
"Nice to Have" Features

- Built-in compliance checklists for ensuring nothing is missed in the review process.
- Workflow management to track the progress of KYC reviews.

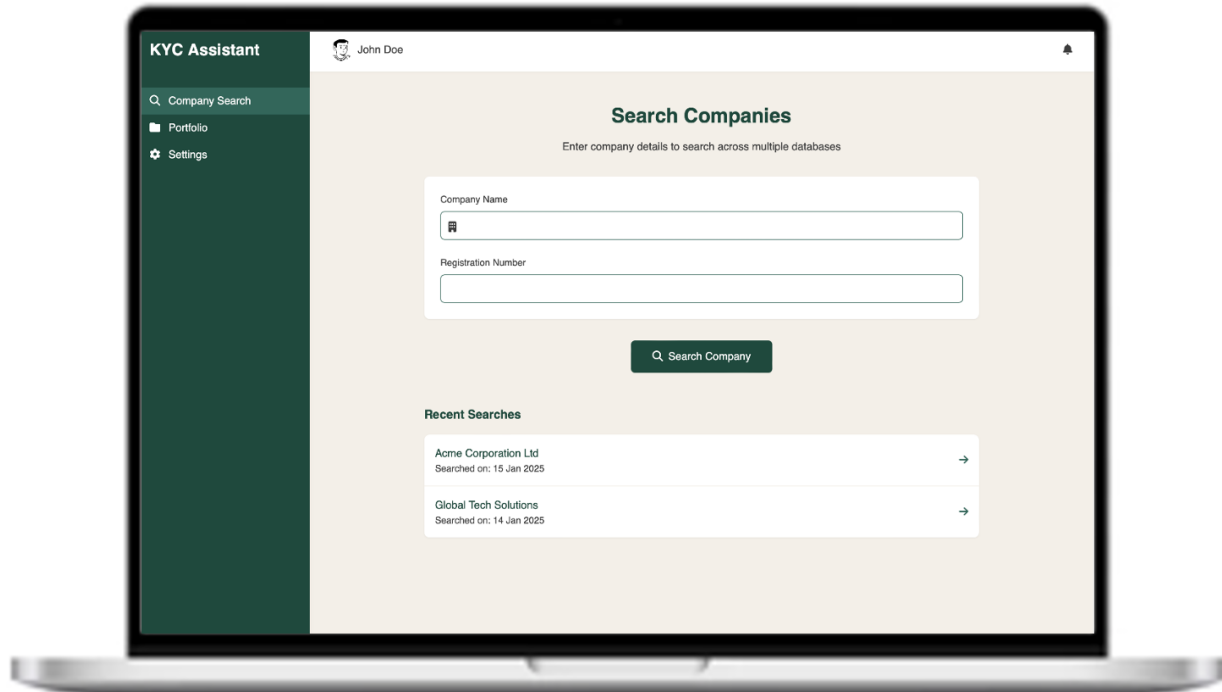
Appendix B – Product Roadmap

	Week 1			Week 2			Week 3			Week 4		
	<i>Planning</i>			<i>Sprint 1</i>			<i>Sprint 2</i>			<i>Sprint 3</i>		
User interviews & observing processes												
Competitor analysis												
MVP feature design												
Prototype wireframes												
MVP architecture												
Key UI interfaces												
User testing												
Add data storage												
Front end improvements												
User testing												
Final release												

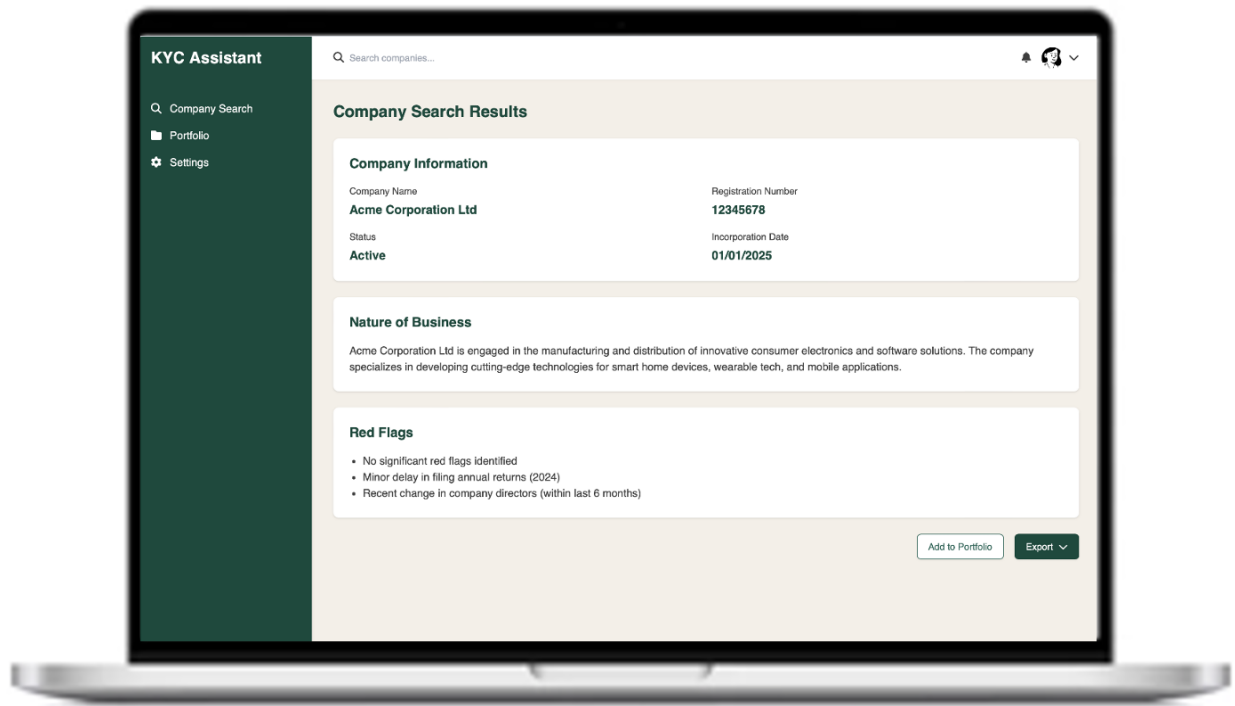
Appendix C – High-fidelity wireframes



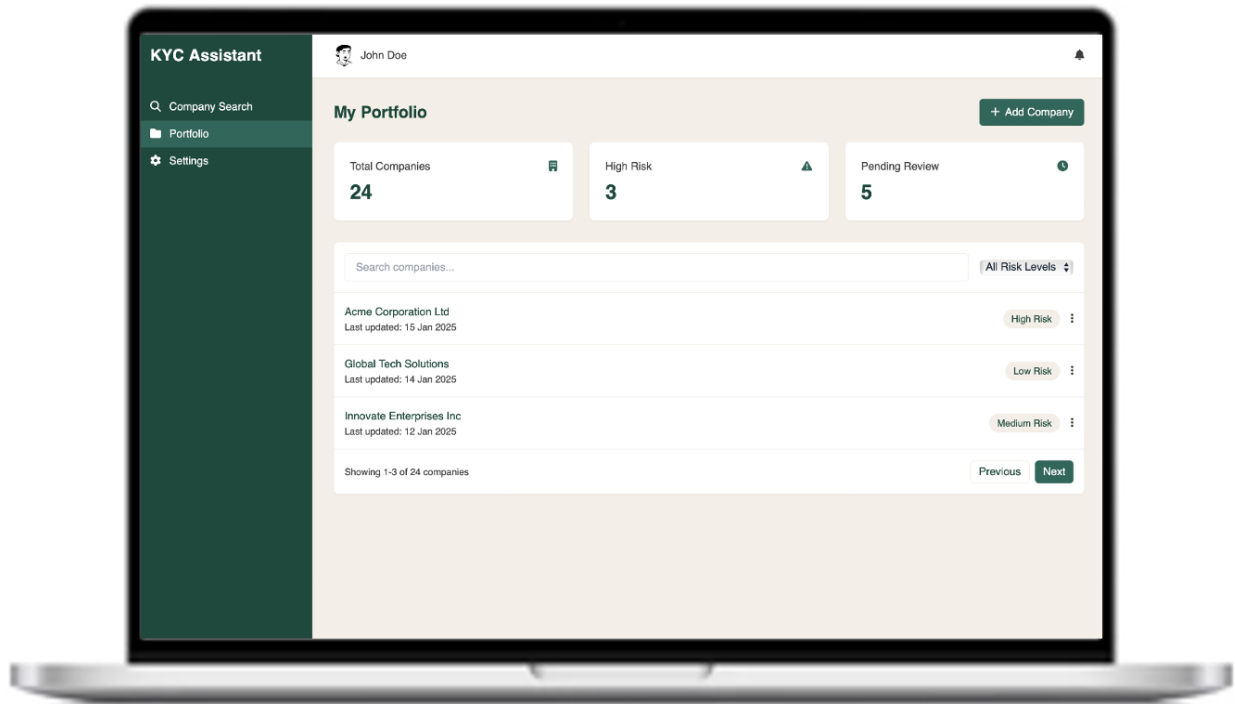
Login page – User story ID US6



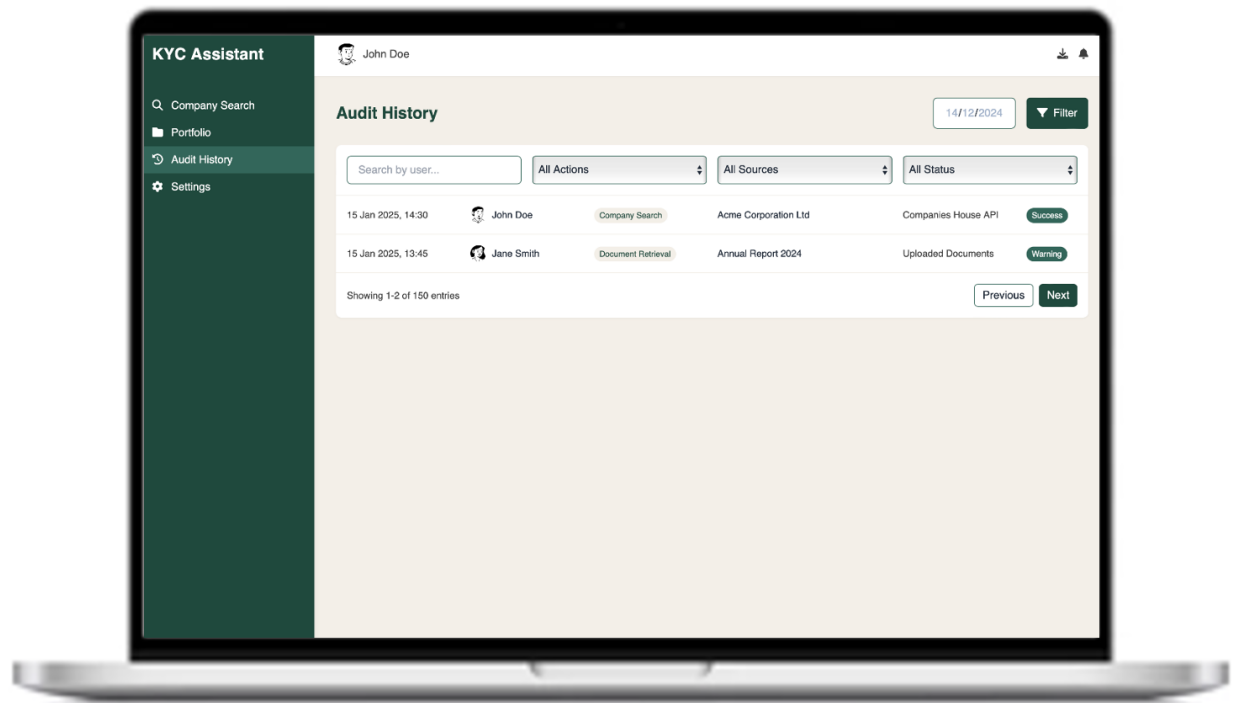
Company search and search history page – User story ID US1 and US9



Company details and export page – User story ID US2 and US3



Portfolio page – User story ID US5



Audit history page – User story ID US7

Appendix D – Detailed demo feedback on Prototypes

The below table details the feedback received on the prototypes over the sprints, plus the next steps required to address that feedback.

Page	Feedback	Next steps
Login page	The options to log in with Google and Microsoft accounts to be removed for this sprint and have been added to the product backlog for future iterations.	Product Backlog update: Add Google and Microsoft account log in functionality to product backlog.
	The “Sign up” option is present on the wireframe but could be more obvious.	Product Backlog update: Add “Sign up” button beneath “Sign in” modal. Create sign in page matching the design of the login page.
	Ensure username and password validation.	Product Backlog update:

		Add username and password validation to the backend.
Search page	User profile could be moved to the bottom left of the screen to neaten the design and increase vertical space.	Product Backlog update: Move the user profile photo and name to the bottom left of the screen.
	It would be useful to be able to clear recent searches to remove clutter.	Product Backlog update: Add a “clear” button to the recent searches section, which should clear the historical search data associated with this account.
Portfolio page	It would be useful to have additional filters, not just “risk level”, such as “date added” and “date incorporated”.	Product Backlog update: Add additional tags to the company data, and additional filter options to the top of the table.
	It would be useful to be able to see more companies on the portfolio page, not just three.	Product Backlog update: Ensure the portfolio page renders 10 companies, or the full portfolio list if the number of saved companies is fewer than 10.
	It would be useful to get an overview of all my portfolio companies on a single export.	Product Backlog update: Add function to export full list of portfolio companies as a CSV file, with a modal that allows you to select the preferred metadata to export (e.g. names, incorporation dates, etc.).
Company details page	KYC analysts would like to see additional information on the details page including registered office address, company type, Directors and the nature of business (SIC ⁷) code.	Product Backlog update: Extract these data points from the UK Companies House API, and display on the page.
	It would be useful to see if the selected company has any connecting data with any other company in my portfolio, such as a shared UBO, Director, or registered address.	Product Backlog update: Cross-reference the selected company with companies in the user’s portfolio for identical data points, and display on the Company details page.
Other	Over the course of sprint 1, it was identified that a search results page (sitting between the company search page and the company details page, where the user can select the company they are interested in) would also need to be designed as a step between searching for the company and showing the company details.	Product Backlog update: Create search results wireframe and page in next sprint.

⁷ Standard Industrial Classification (SIC) code details a company's primary industry.

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